

BRIAN BROST

Statistics + Spatial Analysis + Data Visualization

Data scientist, applied statistician, and spatial analyst with 15+ years of experience finding the signal in a noisy world. Highly skilled in advanced analytical techniques—including contemporary Bayesian hierarchical models—that turn complex (and often messy) data into discovery. Passionate about conveying stories through high-quality graphics and cartographic design. Motivated by challenging problems and a desire to leave the world a little better than I found it.

EDUCATION

2017
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2012

Ph.D., Ecology

Colorado State University

Fort Collins, CO

- Research topic: Developing "boutique" hierarchical spatio-temporal models for complex ecological processes
- Dissertation: *Statistical models for animal telemetry data with applications to harbor seals in the Gulf of Alaska* [PDF](#)
- Advisor: Mevin Hooten

2010
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2007

M.S., Conservation Biology

Northern Arizona University

Flagstaff, AZ

- Research topic: Conservation planning strategies for global climate change
- Thesis: *Use of land facets to design conservation corridors: Conserving the arenas, not the actors* [PDF](#)
- Advisor: Paul Beier

2009
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2007

Graduate Certificate in Applied Statistics

Northern Arizona University

Flagstaff, AZ

- Coursework: regression analysis, time series analysis, quantitative biology, multivariate statistics, applied sampling, applied community analysis, and ecological modeling

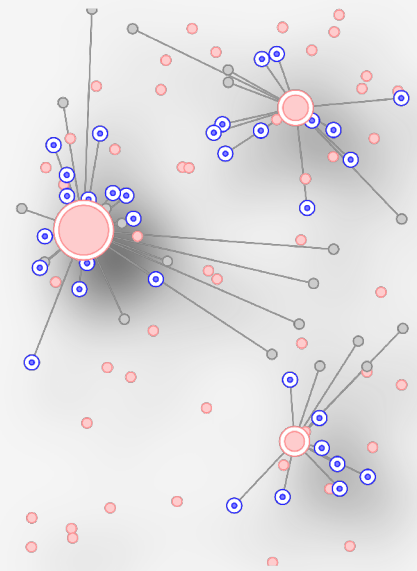
2002
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1999

B.S., Zoology and Conservation Biology

University of Wisconsin-Madison

Madison, WI

- Senior thesis: Characteristics of wolf packs that depredate livestock in Wisconsin, USA



CONTACT

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[scholar.google.com/...](https://scholar.google.com/)

TECHNICAL SKILLS

Tools: R, Stan, NIMBLE, GitHub, Quarto, Markdown, $\text{\LaTeX}$, Adobe Creative Suite, Python, SQL, Shiny, basic HTML & CSS

Statistics: Bayesian hierarchical models, GLMMs, GAMs, HMMs, spatio-temporal models, machine learning, contemporary non-parametrics, frequentist statistics and maximum likelihood estimation

GIS: ArcGIS, QGIS, R (e.g., packages *sf* and *terra*)



PROFESSIONAL EXPERIENCE

2025
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2017

Statistician (Biology)

National Oceanic and Atmospheric Administration

📍 Durango, CO (Remote)

- Led quantitative ecological research and supported a team of ~20 biologists with statistical modeling, spatial analysis, study design, and data management expertise.
- Developed Bayesian hierarchical models for enhanced inference, addressing complexities like measurement error, imperfect observations, missing data, and autocorrelation.
- Analyzed monitoring data to estimate the abundance, survival, habitat use, and distribution of marine mammals.
- Led efforts to estimate marine mammal mortalities incidental to commercial fishing, improving inference and uncertainty accounting.
- Managed a research program investigating movements of northern fur seals using a large instrument array distributed across multiple remote islands.
- Assisted in interpreting complex statistical methodologies, ensuring clarity in technical writing and graphical outputs for publications.

2017
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2012

Graduate Research Assistant

Colorado State University

📍 Fort Collins, CO

- Designed and conducted original research on the spatial ecology of harbor seals in Alaska.
- Developed "boutique" Bayesian spatio-temporal models, including custom software and bespoke MCMC algorithms for model implementation.
- Created statistical methodologies for analyzing biotelemetry data while accounting for locational error, missing data, autocorrelation, and movement constraints.
- Applied novel Bayesian nonparametrics to estimate the location of harbor seal haul-out sites, integrating multiple data sources.

2012
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2010

Biometrician

Great Lakes Indian Fish & Wildlife Commission

📍 Oshkosh, WI

- Provided statistical support to a team of wildlife ecologists, fisheries biologists, and botanists.
- Assessed factors affecting nest site selection by American marten, quantified resource selection by elk, and modeled mercury concentration in walleye to establish consumption guidelines (among other projects).
- Developed and managed PostgreSQL databases for efficient data storage and retrieval.

2010
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2007

Graduate Research Assistant

Northern Arizona University

📍 Flagstaff, AZ

- Developed an innovative method for modeling conservation corridors utilizing topography instead of factors like vegetation that are sensitive to climate.
- Conducted a formal evaluation of corridor modeling approaches to identify best practices.

2007
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2003

Biological Research Technician

Multiple agencies and non-governmental organizations

📍 CA, MT, NV, WY, WA

Participated in various plant- and animal-related fieldwork projects. Duties included:

- Supervised field operations, established sampling priorities, developed field protocols, conducted quality control, and interviewed/hired/trained > 20 field technicians (as Project Coordinator and Crew Leader).
- Conducted wildlife surveys, collected genetic samples using non-invasive techniques, captured/handled animals, studied animal behavior, and mapped vegetative and abiotic land cover (as Crew Member).



PUBLICATIONS

- In prep. • Sterling, J.T., and B.M. Brost. Summer resource limitation as a likely mechanism impacting northern fur seal (*Callorhinus ursinus*) recovery.
- In prep. • Brost, B.M., and N. Young. Hierarchical models improve inference concerning marine mammal mortalities incidental to commercial fishing activities.
- In review • Zeppelin, T.K., B.M. Brost, D.S. Johnson, C.M. Kurle, B.A. Güell, C. Keller, M.T. Williams, and A.E. York. Determining cryptic life history milestones using isotopic markers in northern fur seal (*Callorhinus ursinus*) vibrissae.
- 2022 • Conn, P.B., J.M. Ver Hoef, B.T. McClintock, D.S. Johnson, and B. Brost. A GLMM approach for combining multiple relative abundance surfaces. *Methods in Ecology and Evolution* 13:2236–2247. [PDF](#)
- 2022 • Johnson, D.S., B.M. Brost, and M.B. Hooten. Greater than the sum of its parts: computationally flexible Bayesian hierarchical modeling. *Journal of Agricultural, Biological, and Environmental Statistics* 27:382–400. [PDF](#)
- 2021 • Hooten, M.B., D.S. Johnson, and B.M. Brost. Making recursive Bayesian inference accessible. *The American Statistician* 75:185–194. [PDF](#)
- 2020 • Brost, B.M., M.B. Hooten, and R.J. Small. Model-based clustering reveals patterns in central place use of a marine top predator. *Ecosphere* 11(6):e03123. [PDF](#)
- 2019 • Fritz, L., B. Brost, E. Laman, K. Luxa, K. Sweeney, J. Thomason, D. Tollit, W. Walker, and T. Zeppelin. A re-examination of the relationship between Steller sea lion (*Eumetopias jubatus*) diet and population trend using data from the Aleutian Islands. *Canadian Journal of Zoology* 97:1137–1155. [PDF](#)
- 2019 • Zeppelin, T., N. Pelland, J. Sterling, B. Brost, S. Melin, D. Johnson, M. Lea, and R. Ream. Migratory strategies of juvenile northern fur seals (*Callorhinus ursinus*): bridging the gap between pups and adults. *Scientific Reports* 9:13921. [PDF](#)
- 2018 • Brost, B.M., B.A. Mosher, and K.A. Davenport. A model-based solution for observational errors in laboratory studies. *Molecular Ecology Resources* 18:580–589. [PDF](#)
- 2018 • Davenport, K.A., B.A. Mosher, B.M. Brost, D.M. Henderson, N.A. Denkers, A.V. Nalls, E. McNulty, C.K. Mathiason, and E.A. Hoover. Assessment of chronic wasting disease prion shedding in deer saliva with occupancy modeling. *Journal of Clinical Microbiology* 56:e01243–17. [PDF](#)
- 2017 • Brost, B.M., M.B. Hooten, and R.J. Small. Leveraging constraints and biotelemetry data to pinpoint repetitively used spatial features. *Ecology* 98:12–20. [PDF](#)
- 2017 • Hefley, T.J., K.M. Broms, B.M. Brost, F.E. Buderman, S.L. Kay, H.R. Scharf, J.R. Tipton, P.J. Williams, and M.B. Hooten. The basis function approach for modeling autocorrelation in ecological data. *Ecology* 98:632–646. [PDF](#)

- 2017 • Hefley, T.J., B.M. Brost, and M.B. Hooten. Bias correction of bounded location errors in presence-only data. *Methods in Ecology and Evolution* **8**:1566–1573. [PDF](#)
- 2017 • Small, R.J., B. Brost, M. Hooten, M. Castellote, and J. Mondragon. Potential for spatial displacement of Cook Inlet beluga whales by anthropogenic noise in critical habitat. *Endangered Species Research* **32**:43–57. [PDF](#)
- 2016 • Hooten, M.B., F.E. Buderman, B.M. Brost, E.M. Hanks, and J.S. Ivan. Hierarchical animal movement models for population-level inference. *Environmetrics* **27**:322–333. [PDF](#)
- 2015 • Brost, B.M., M.B. Hooten, E.M. Hanks, and R.J. Small. Animal movement constraints improve resource selection inference in the presence of telemetry error. *Ecology* **96**:2590–2597. [PDF](#)
- 2012 • Brost, B.M., and P. Beier. Comparing linkage designs based on land facets to linkage designs based on focal species. *PLoS One* **7**:e48965. [PDF](#)
- 2012 • Brost, B.M., and P. Beier. Use of land facets to design linkages for climate change. *Ecological Applications* **22**:87–103. [PDF](#)
- 2010 • Beier, P., and B. Brost. Use of land facets to plan for climate change: Conserving the arenas, not the actors. *Conservation Biology* **24**:701–710. [PDF](#)
- 2010 • McMaster, M.A., A. Thode, B. Brost, M. Williamson, E. Aumack, D. Mertz. Changes in vegetation and fuels due to the Warm Fire on the Kaibab National Forest. *Joint Fire Science Program Research Project Reports* 34. [PDF](#)
- 2004 • Wydeven, A.P., A. Treves, B. Brost, and J.E. Wiedenhoef. Characteristics of wolf packs in Wisconsin: Identification of traits influencing depredation. In *People and Predators: From Conflict to Coexistence*, eds. N. Fascione, A. Delach, and M. E. Smith. Island Press, Washington, D.C. [PDF](#)